Sentiment Analysis Towards Zohran Mamdani in YouTube Comments

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***Abstract*—Online activity is becoming increasingly important in political campaigns. This paper further fine-tunes a sentiment analysis fine-tuned BERT model on a small sample of YouTube comments about NYC Mayor-Elect Zohran Mamdani. Using Data Augmentation, the fine-tuned model was improved. This model was applied to many more YouTube comments from YouTube channels CNN, Fox News, and Breaking Points, and sentiment towards Mamdani was tracked over time. No clear trend over time was found though sentiment on Fox News was consistently lower than the other sources.**

***Index Terms*—Fine-Tune, BERT, Political Sentiment, Sentiment Analysis, Machine Learning, Natural Language Processing, Social Science**

I. Introduction

A. Background

Zohran Mamdani made history on November 4th 2025 by becoming the first Muslim, first Indian mayor of New York City, as well as being the youngest NYC mayor elected since 1892. To some, his victory was seen as a complete surprise but to others it was inevitable after disappointments from the establishment candidate. In addition to policy and traditional outreach, online presence and sentiment is an influential factor in a political candidate's success. Understanding the online sentiment towards a candidate can be an indicator of their success on election day.

The motivation of this project was to correctly classify sentiment of a large number of comments towards Mamdani, and track the sentiment towards him from the beginning of his campaign, which launched on October 24th 2024, until the day before the New York Mayoral election, November 3rd 2025. In order to do so, classifying sentiment of comments was to be done using a fine-tuned language model to get a more accurate classification.

The final goal of the project was to fine-tune a BERT model to predict sentiment towards Mamdani, and apply it to track sentiment towards Mamdani in YouTube comments, specifically under news channels, over the length of his campaign.

B. Literature Review

A lot of existing literature about sentiment analysis, especially involving political attitudes, has been focused on Twitter and Reddit posts. However, both platforms have tightened their (formerly free) API access, making YouTube a more novel approach, especially with fine-tuning.

A Hungarian study [1] performed fine-tuning on a BERT model with manually classified they use Precision, Recall, and Accuracy to test different epochs and hyperparameters. They also performed emotion analysis with emotions "Joy, Anger, Sadness, Shame, Guilt, Surprise, and Fear."

Another study used another BERT HuggingFace model to predict the sentiment of tweets and reddit posts about the COVID vaccine. [2] They had a team and were able to feed 3,600 posts into fine-tuning that were manually labeled. They used back-translation to augment the dataset to account for more tweets. They graphed polarity when looking at tweets about the vaccine.

II. Methods

A. Fine-Tuning

The first step in the project was obtaining YouTube comments that could be fine-tuned with a BERT model. Due to time, resource, and API constraints, 239 comments were retrieved and manually classified, with the classification specifically being the sentiment towards Zohran Mamdani, as either 1 (positive) or 0 (negative). Only comments that *explicitly included* Mamdani's name (either "zohran" or "mamdani") were chosen.

These YouTube comments were obtained from a variety of sources in order to receive a high diversity of comments. These videos included the Final New York Mayoral debate, interviews with Mamdani, and one of his campaign ads. Extra attention was paid to ensure there were at least 100 negative and 100 positive comments, so negative comments were chosen from news clips as well as YouTube videos from popular political figures (Megyn Kelly, Dave Rubin).

Due to the small dataset size, the HuggingFace model `distilbert-base-uncased-finetuned-sst-2-english` was chosen.

This model is already fine-tuned; it utilizes a base BERT model and fine-tuned for sentiment analysis on movie reviews. This is an ideal model because the sentiment analysis task is already learned, and allows the small set of YouTube comments to further fine tune with specific language about the political candidate.

To further address the issue of very few labeled examples, a few methods were put in place:

- Freeze Base Model Parameters

- Data Augmentation - Random Character Swap
- Data Augmentation - Random Insert / Delete

Data Augmentation roughly multiplied the size of the dataset by 6 times, in which the `textattack` library was used for random character swapping, deleting, and inserting.

Inputs: 1144 Labeled Training Examples (including augmented examples).

Output: Fine-Tuned Model

Evaluation was done on this further fine-tuned model with all adjustments made, along with a model without the data augmentation, and the original base model without extra fine-tuning. These were evaluated using a validation set of 48 examples, using accuracy and f1-score, to see if improvements were being made.

B. Application of Model and Analysis

The next part of the project was the application of the fine-tuned model. Due to YouTube's API quota, application of the fine-tuned model was limited to three news sources: CNN, Fox News, and Breaking Points. CNN and Fox News are extremely popular news sources, and Breaking Points is an independent news source with politically diverse viewership.

The YouTube API allows for the top 100 comments from a video to be retrieved. For each channel, these top 100 comments were retrieved for every video uploaded between October 24th 2024 to November 3rd 2025. Using only the comments mentioning "zohran" or "mamdani" the best fine-tuned model predicted their sentiment.

At the end there were 5,134 comments total. The data was aggregated by day, week, and month, for any noticeable trends over time. Due to some instability in the number of comments week by week, a daily rolling average over 7 days was calculated to smooth the noisy examples.

III. Results

A. Model Performance

To ensure that the model was not deteriorating with the extra examples, or overfitting, three models were compared.

TABLE I
MODEL COMPARISON

Validation Scores			
Model	***Loss***	***Accuracy***	***F1-Score***
BERT SST-2 Pre-Fine-Tuned	1.096	0.729	0.730
Fine-Tuned	0.482	0.771	0.768
with Data Augmentation	0.462	0.813	0.810

Clearly, the fine-tuned model with Data Augmentation outperforms the other models in each metric, implying it would be better to use for further classification.

Examining the changes in prediction from the base model to the final model, there were clearly some improvements.

Example a: "there was a lot of planning and effort in the failed 1993 wtc bombing. but with mamdani as mayor, they'll just be able to walk right in to the stock exchange - no questions asked."

Example b: "pathetic cuomo couldn't even take the time to shake the hands of his opponents. runs to the moderators and then scurries away. zohran and curtis did immediately."

Example	***True***	***Base Model Pred***	***FT with Data Aug. Pred***
a	1	0	1
b	0	1	0

Whether or not it was the "intention" of the model, the fine-tuned model correctly predict sthe sentiment towards the Mamdani specifically where the base model does not.

B. Sentiment Analysis

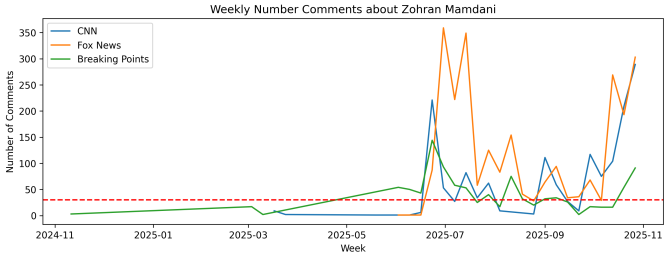


Fig. 1. Weekly Number of Comments about Zohran Mamdani

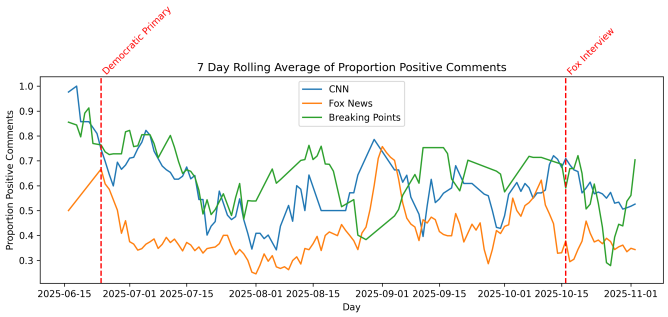


Fig. 2. Rolling Average of Sentiment about Zohran Mamdani

The number of comments by week in Figure 1 help explain some of the variability in the sentiments. Some weeks have as many as 200+ comments about Mamdani, while others have less than 10. This provides context for the sentiment graph, and the red-line separates where the sentiment proportions may be more stable.

Figure 2 shows the final sentiments across the three channels and how they change over time. The 7 day rolling averages partially solves the instability issue, however even so some periods still have very few comments.

Unsurprisingly, Fox News has consistently lower sentiment than the other two news sources. After some investigation the days with the lowest sentiments have videos very deliberately negative towards Mamdani, or that week has very few comments in general. Also very clearly Breaking Points has had comments about Mamdani from as early as October 2024, which makes sense as their populist believes line up with an non-establishment candidate. It also makes sense that their

sentiments are higher in general compared to the mainstream media sources.

The trend over time seems fairly noisy but interestingly Fox News seems to have more positive sentiments towards the last part of the campaign. This may be due to supporters disagreeing with the content of the video and expressing support in the comments.

One thing to note is that the model did only perform with a 80% accuracy, so the comments are not all perfectly classified. For example, there were multiple examples of comments that were favorable towards Zohran Mamdani, but generally had a "negative" sentiment, though this was improved with fine-tuning.

Example c: "the amount of hostility zohran mamdani has received from both the right and the left because he wants to help working class american people who are struggling is disgusting!! the billionaires are supposed to pick our leaders?? the voters don't get to choose their leaders in the usa?"

Example c was predicted as 0 even though they are clearly supporting Mamdani.

IV. Conclusion

In conclusion, there is value in fine-tuning a model on the specific task of political sentiment towards a specific candidate, especially in the online discussion space with sarcasm, slang, and expressing support in a "negative" comment. Improvements can be made by feeding in classified examples, along with natural language processing techniques like Data Augmentation. After using the model on comments from CNN, Fox News, and Breaking Points, there doesn't seem to be a clear trend with the noisy data, however, comparing each channel it is clear that there is a difference in sentiments, with Fox News having the lowest sentiment in general, but with sentiments generally being higher closer to election day.

A lot of improvements can be made to this project for more insightful and accurate results, particularly by obtaining more training samples as well as more comments to use afterwards during the sentiment analysis. Further work might involve obtaining not only YouTube comments but also posts on other social media platforms.

References

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